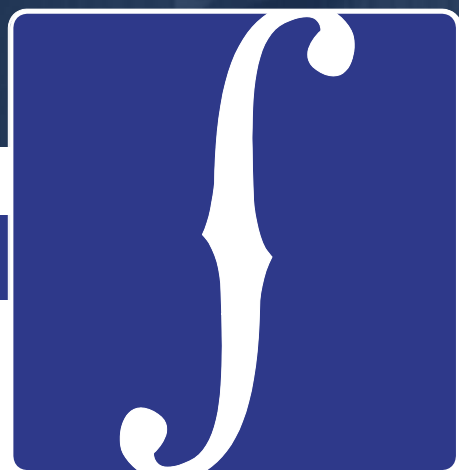




CELLO

Identifying HARPed Loans

February 2016

Key Differences in Streamline Refi, HARP 1.0 and HARP 2.0 Programs

	HARP 2.0	HARP 1.0	Streamline Refi
Period	Since November 2011	April 2009 – October 2011	Pre-HARP 1.0
Appraisal	Mostly Automated	Manual/automated appraisal but manual appraisal required in most cases	Manual appraisal
Income & Asset documentation	Reduced requirement in September 2012--same servicers are required to verbally verify a borrower's income; New lenders are required to have documentation	Need documentation; in some cases need two year history of income	Stated Income/employment
Reps & Warrants	Further relief of reps & warrants for original loan in September 2012; Relief of new loan reps & warrants risk with 12 month clean pay following origination	Seller retains all the original reps & warrants regarding value condition and marketability; no additional relief on the new loan	Any previous rep & warrant relief on the original loan does not apply to the new loan
LTV restriction	Extended to above 125% LTV in October 2011	LTV ceiling at 105% when announced in March 2009; Raised ceiling from 105% to 125% in July 2009	under 95%-100%
MI	MI transfers made easier	No new MI required but transfer of MI is cumbersome	New MI required
LLPA	Lowered LLPA to 75bp max	Standard	Standard

1

Importance of identifying HARP loans

Loans that have been refinanced through the Home Affordable Refinance Program (HARP) generally prepay slower than other refinance loans that have undergone full underwriting. As a streamline refinance program, HARP generally made only minimal requirements of borrowers, and such borrowers may subsequently have trouble qualifying for regular refinances.

2

Existing algorithms for identifying HARPed Loans

1. Prior to August 2013:
 - This is the period prior to Freddie Mac providing loan-level credit information for fully underwritten mortgages
 - MI matches were used to identify potential HARPed loans with OLTV > 80% and loan purpose as Refi-NCO (No Cash-out)
 - Although HARP targets with LTVs > 80%, loans with lower OLTVs are counted as part of HARP because the Agencies essentially suspended their streamline refi programs after the financial crisis and currently rely on the HARP program for all streamline processing
2. After August 2013:
 - New Freddie Mac and Fannie Mae loan-level credit information are used to identify 30-year fixed –rate HARP loans by mapping the credit dataset to the standard database without restrictions on the LTV ratio
 - The credit data set has about a one year lag, which means that recently originated loans cannot be identified as being HARPed or not using the current algorithm

Three Step Mapping

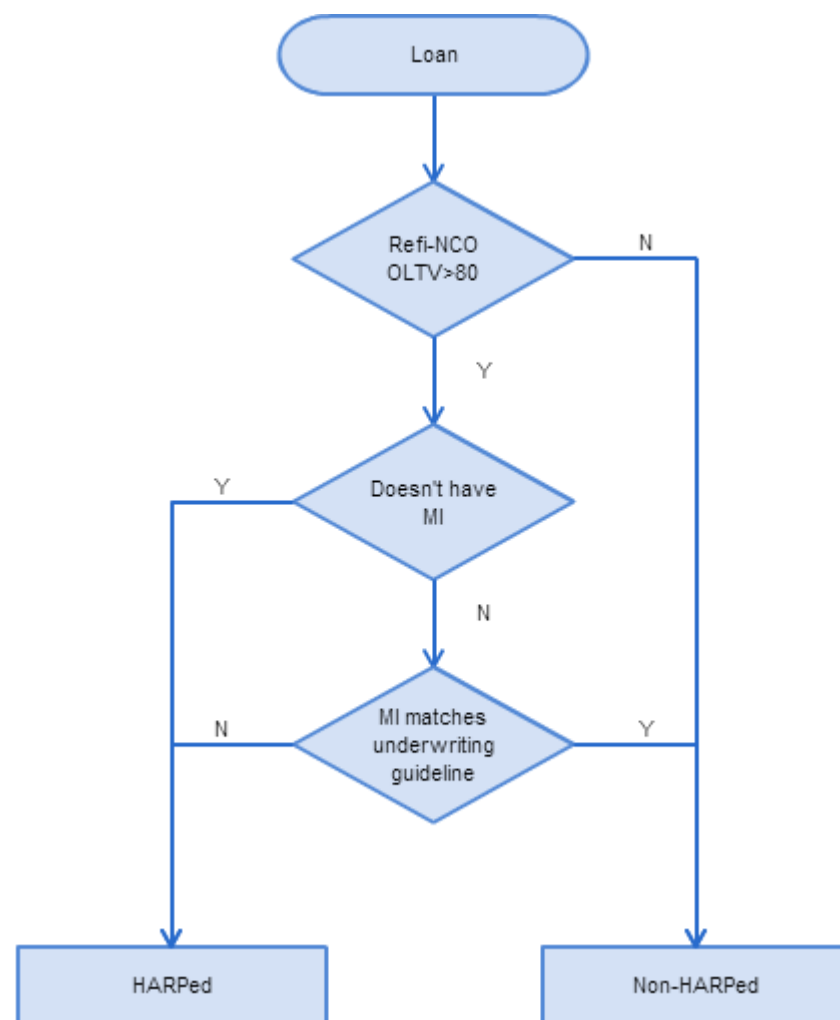
Step 1: The loan should be originated after June 2009 as a refi-NCO loan, with Origination LTV >80%. Otherwise, it's Non-HARPed.

Step 2: If the loan doesn't have MI, we match it as HARPed loan;
If not, go to step 3.

Step 3: For the loans that potentially carry over MI check whether MI matches current underwriting guidelines. If there's no match, then it's a HARPed loan.

Cons

1. Cannot match loans with OLTV equal or lower than 80%.
2. Highly sensitive to the correctness of MI underwriting guidelines.

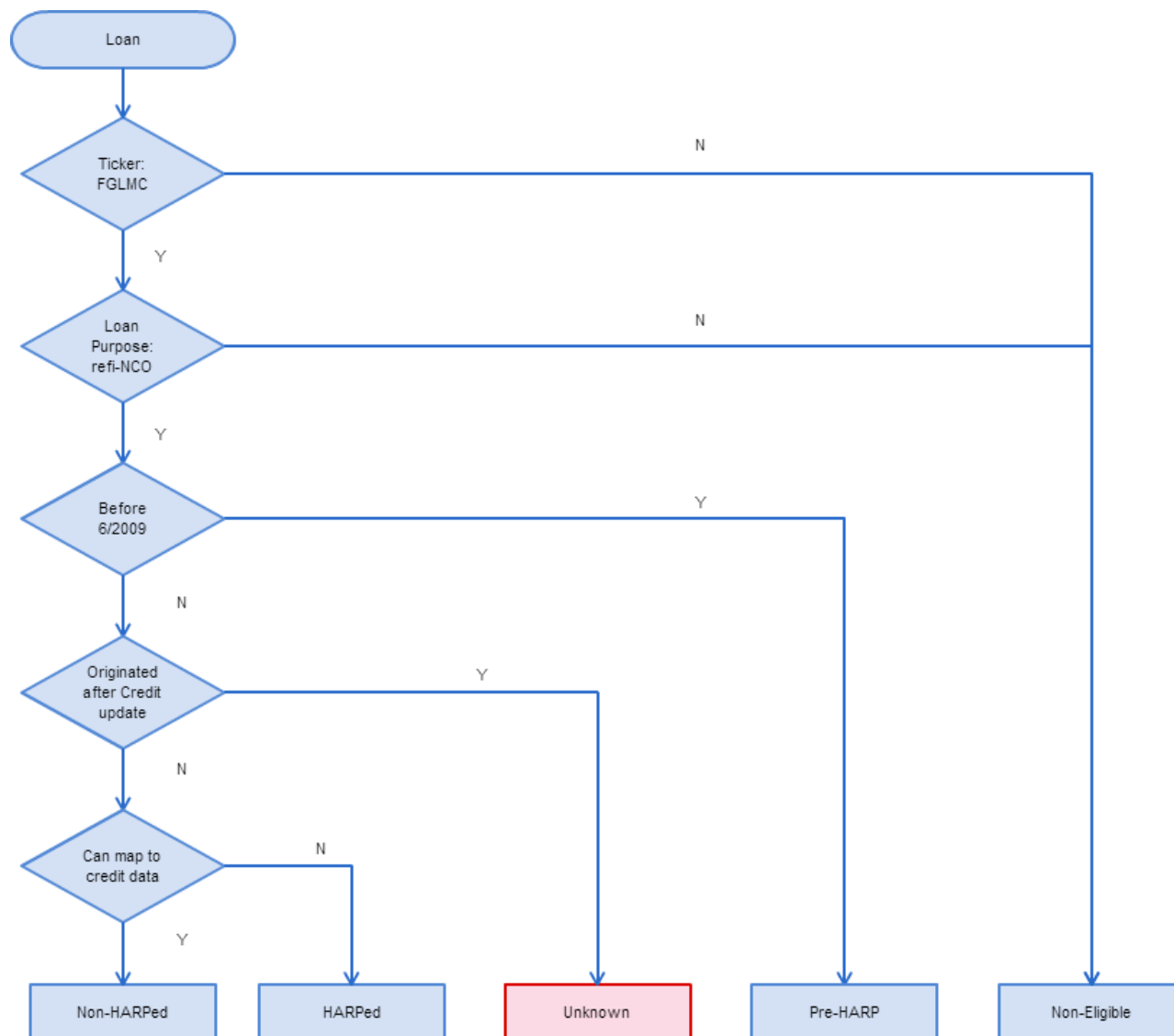


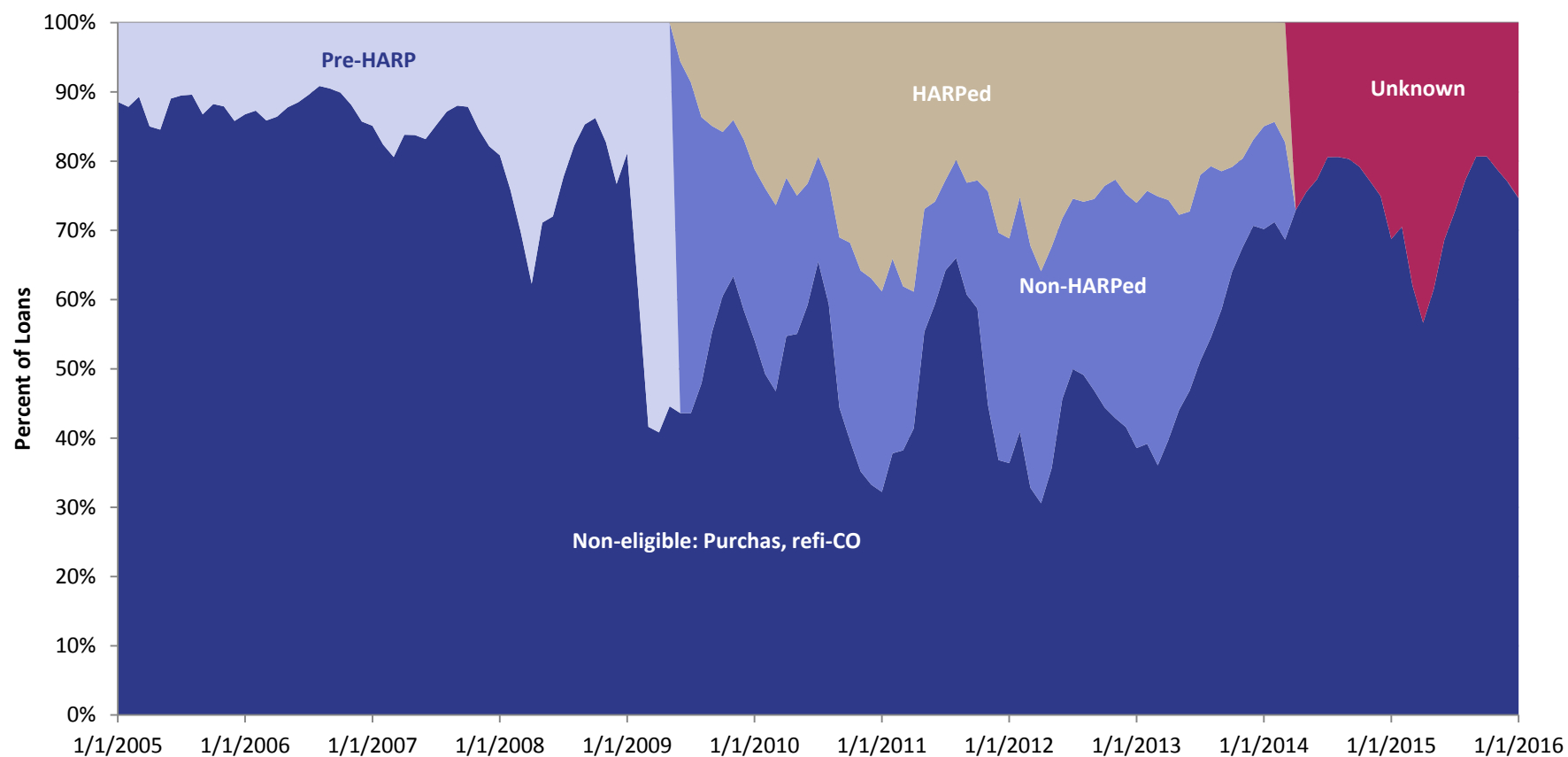
Credit Data Mapping

- No restrictions on OLTV > 80%.
- More accurate as there is no dependency on MI underwriting criteria.

Cons

- The credit data set has about a one year lag, meaning we don't have the most recent origination data. Consequently, we have to mark the HARP status as unknown for recently (within the last year) originated loans.

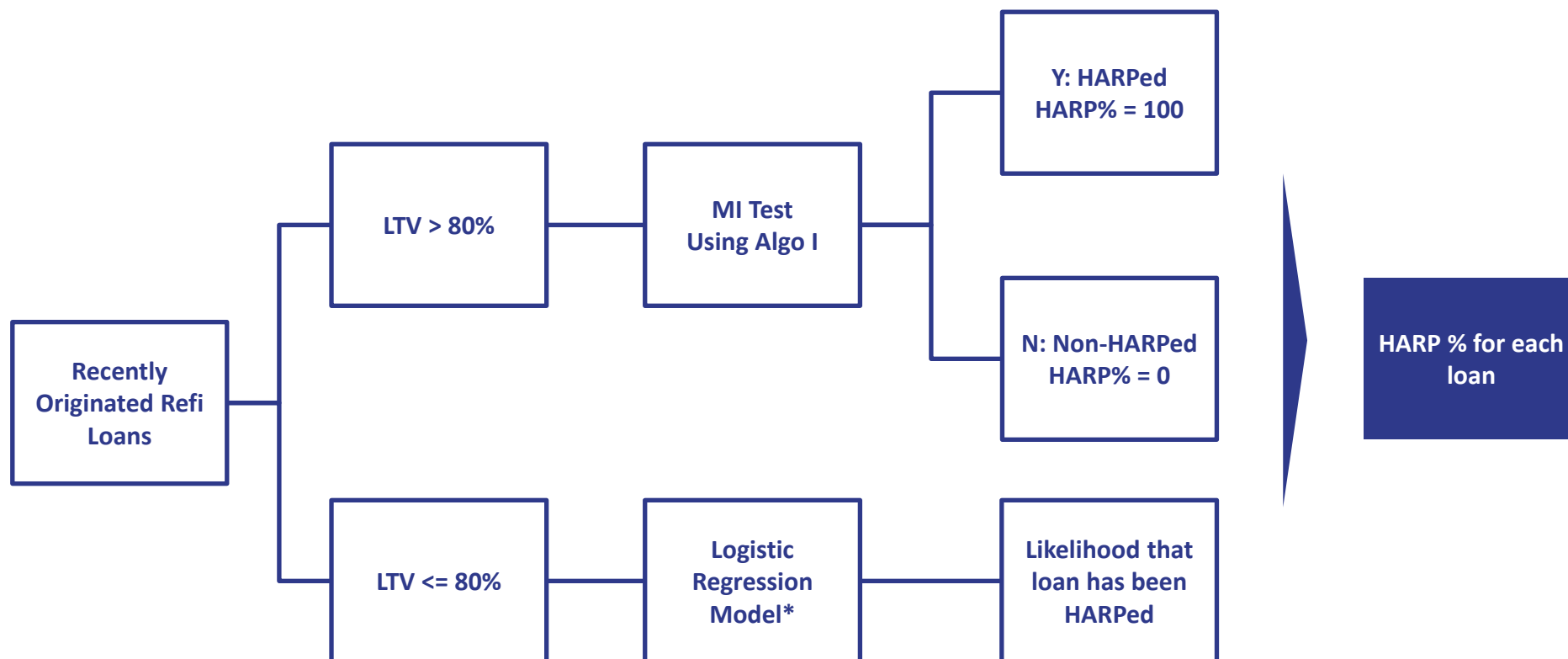




The blue area can be consider as non-HARPed loans; brown area is HARPed loans percentage.

Red area is the loans originated within the most recent year. The credit dataset has a gap on this area.

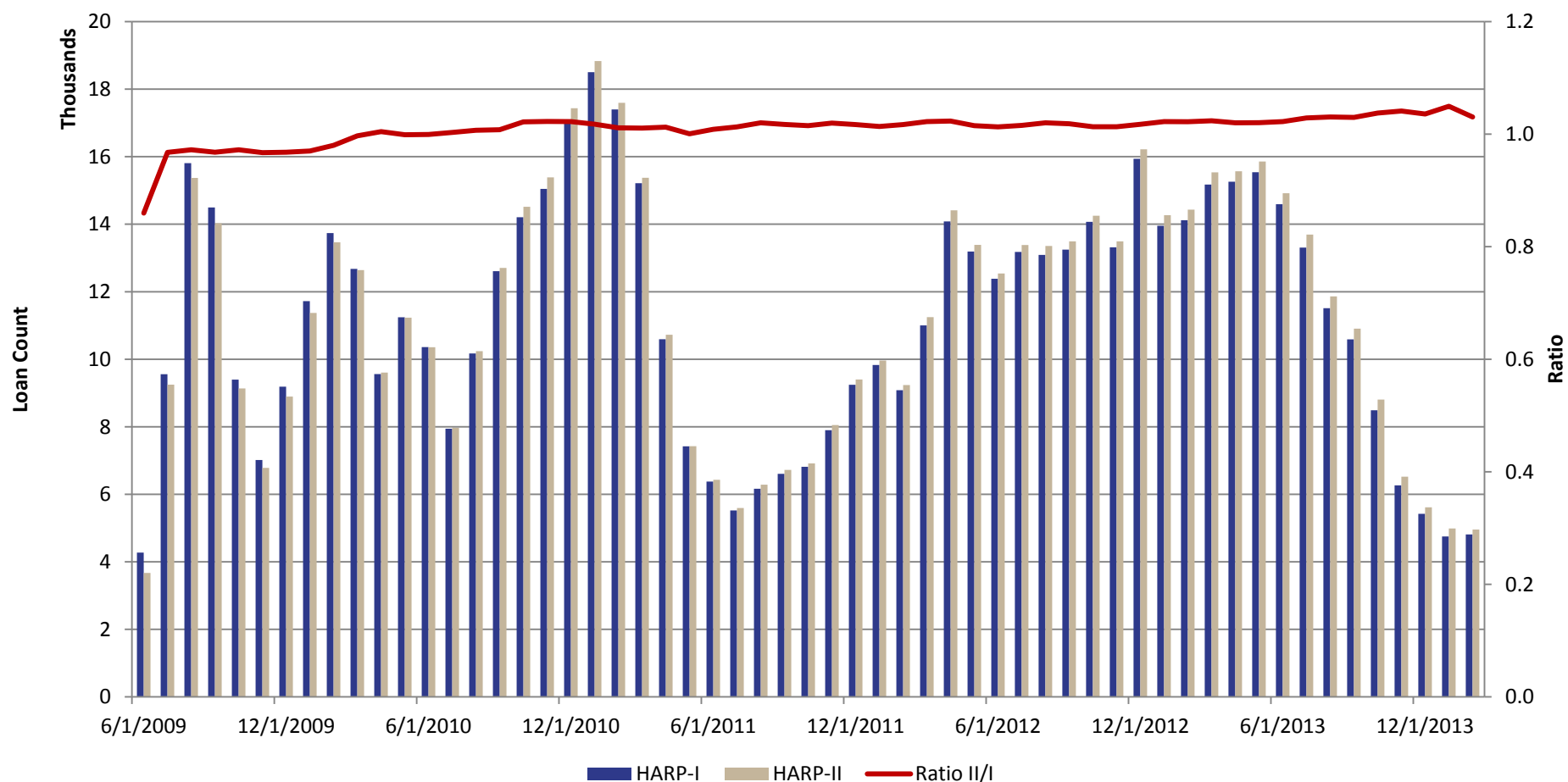
Need to improve the mapping for the red area.



We estimate the likelihood that the loan is originated from the HARP program for each loan.

How good is the MI matching algorithm at identifying HARP loans with LTV > 80%? We backtested the algorithm on HARP loans identified by the Credit Dataset matching algorithm. The HARP mapping numbers from them are fairly close to each other. Consequently, there is no need to apply an adjustment factor when running the MI algorithm on recently originated LTV > 80% loans.

Ratio Algo II/Algo I (LTV > 80%)



Logistic Regression Model for the HARPed Percentage

For loans with LTV ≤ 80%, and originated after the most recent available credit data set, we cannot use neither Algorithm I or Algorithm II to estimate the percentage of loans that are HARPed. In order to solve the problem, we introduce a logistic regression model based on loan characteristics and global estimates of the refinanceable universe. The algorithm is calibrated based on the HARPed vs non-HARPed breakout generated by the Credit Dataset matching algorithm.

$$\text{HARPed Percentage} \sim \text{Credit Score} + \text{SATO} + \text{OrigLoanSize} + \text{Refi \%} + \text{HARP Eligible \%}$$

Loan Characteristics:

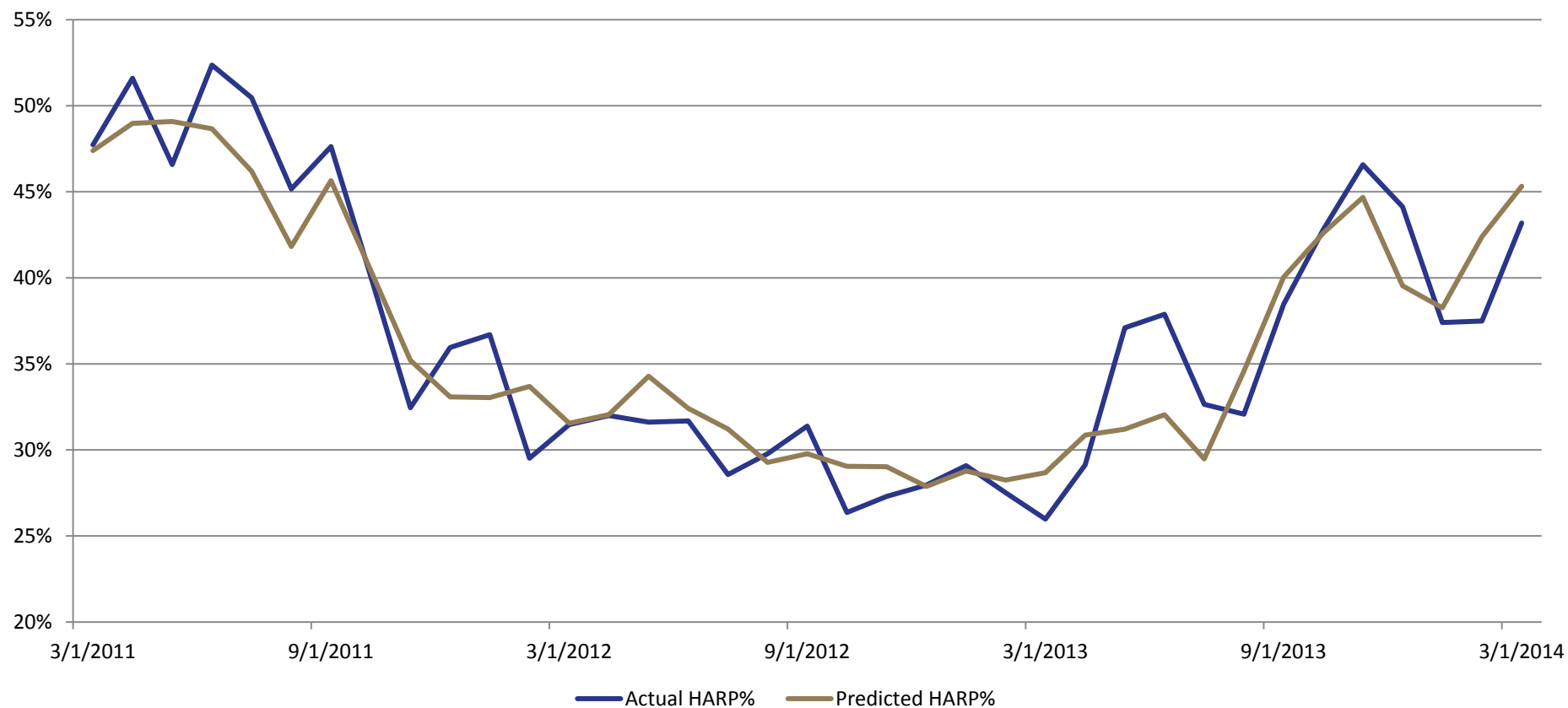
1. Credit Score: The lower the credit score, the higher probability the loan is HARPed.
2. SATO: Borrowers with higher spreads normally have higher HARPed probabilities
3. OrigLoanSize: A lower loan size is associated with a higher chance that a loan is HARPed.

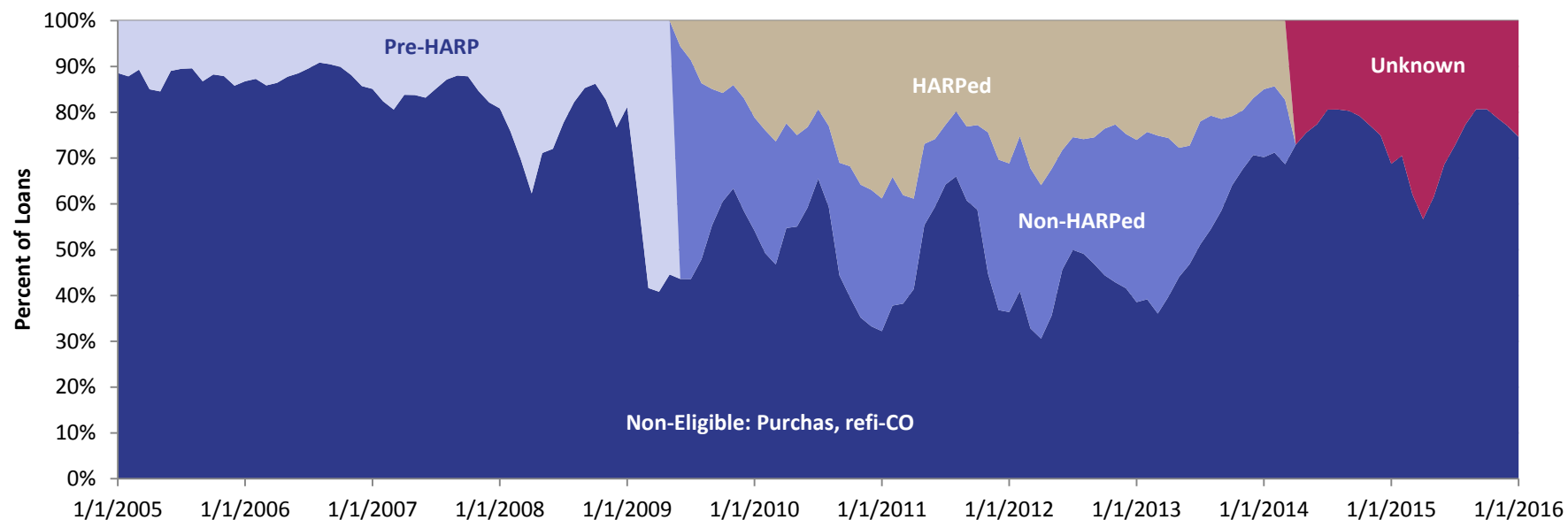
Refinanceable Universe Indexes:

1. Refinanceable Percentage: The percentage of all loans that have > 50bps incentive to refinance. When the Refi % is higher, the HARPed % of all refinance loans is lower (HARP-eligible loans prepay less dynamically than non-HARP-eligible loans) .
2. HARP Eligible Percentage: Higher ratios imply that the probability of a loans being HARPed is higher.

Logistic Regression Model for HARP Percentage

The graph below represents an out-of-sample test of our regression model. The model is trained using 80% of HARPEd vs non-HARPEd dataset, while holding out the remaining 20% as out-of-sample test set.





Using the Enhanced Algorithm

